

Project Initialization and Planning Phase

Date	17 JULY 2026
Team ID	SWTID-2026-4144
Project Title	RIISNG WATERS
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The Proposal report aims to save lives and property in flood-prone regions using machine learning+IOT. It tackles delayed and generic flood warnings by providing hyper-local,real-time predictions. Key features include an ML-based flood prediction model,live sensor data integration,and instant alert via app/SMS to resident and officials.

Project Overview	
Objective	The primary objective is to reduce loss of life and property by implementing an ML-based system that predicts floods using IOT sensor data+IMD weather data,ensuring early and accurate alerts.
Scope	The project covers real-time data collection from IOT sensors,rainfall prediction using ML,location based alert generation,and delivery of alerts through mobile app,SMS,and offline channels like sirens.
Problem Statement	
Description	Current flood warning systems are delayed,generic,and non-location-specific. This leads to panic late evacuations ,property damage ,and loss of life. There is no centralized system to predict floods before they occur using real-time data.

Impact	Solving these issues will result in early evacuations, reduced property damage, minimized loss of life, and improved disaster response efficiency. It will build public trust and save costs for govt and communities.
Proposed Solution	
Approach	Employing machine learning techniques to analyze real-time IOT sensor data + IMD weather data to predict water levels and floods risk. This creates a proactive, location-specific alert system instead of reactive warnings.
Key Features	- Implementation of a machine learning-based flood prediction model using LSTM/Regression.

	- Real-time data collection from IOT sensors deployed in rivers/reservoirs - Hyper-local alerts via mobile app, SMS, whatsapp for residents.
--	---

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU for training ML model & real-time inference	T4 GPU / Intel i5 or above
Memory	RAM for model training, data processing, backend	8 GB - 16 GB
Storage	Disk space for sensor data, weather logs, trained models	1 TB SSD / Cloud storage
Software		

Frameworks	Python frameworks for backend&API	Flask /Django/FastAPI
Libraries	ML+Data handling+visualization libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE used for coding & model training	Jupyter Notebook, pycharm ,VS code
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv